



Solution Requirements

Date	26/06/2026
Team Id	
Project Name	Smart Lender
Maximum Marks	4 Marks

Define Functional Requirement And Non Functional Requirements

Functional Requirement:

Functional requirements describe the specific features and functions that the Smart Lender system must perform to meet user needs, such as accepting applicant details, validating data, predicting loan approval, and displaying results.

Non-Functional Requirement :

Non-functional requirements define the quality attributes and performance standards of the Smart Lender system, such as speed, accuracy, security, reliability, usability, and maintainability.

Functional Requirements:

Requirement Category	Requirement Description	Priority
User Registration & Input	The system shall allow users to enter loan applicant details through a secure web form.	High
Data Validation	The system shall validate all mandatory fields and ensure that the entered data is accurate before processing.	High
Loan Prediction	The system shall use the trained Machine Learning model to predict whether the loan will be approved or rejected.	High
Result Display	The system shall display the loan prediction result instantly in a clear and understandable format.	High
Error Handling	The system shall display appropriate error messages for invalid, incomplete, or incorrect user inputs.	Medium

Requirement Category	Requirement Description	Priority
Model Management	The system shall load the trained Machine Learning model at application startup and use it for prediction requests.	Medium

NON FUNCTIONAL REQUIREMENTS:

NFR Category	Description	Target Metric
Performance	The system should generate the loan prediction quickly after the user submits the application.	Response time $\leq$ 3 seconds
Accuracy	The Machine Learning model should provide reliable loan prediction results.	Prediction accuracy $\geq$ 85%
Usability	The application should have a simple, intuitive, and user-friendly interface.	New users can complete a prediction in less than 2 minutes
Reliability	The system should operate consistently without unexpected failures.	System availability $\geq$ 99% during normal operation
Security	User input data should be processed securely and protected from unauthorized access.	100% input validation and secure data handling
Maintainability	The application should be easy to update, debug, and enhance with new features or models.	New features can be integrated with minimal code changes